

Double groups of gates based Takagi-Sugeno-Kang (DG-TSK) fuzzy system for simultaneous feature selection and rule extraction ☆

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Received 24 June 2022; received in revised form 25 April 2023; accepted 14 June 2023

Available online 27 June 2023

Abstract

Embedded feature selection methods based on fuzzy model have been well studied, in which the gate functions are usually used to distinguish the essential features. Fuzzy rules play a critical role in constructing fuzzy systems. However, few works simultaneously evaluate the quality of fuzzy rules during the fuzzy system based feature selection approaches. This article proposes double groups of gates based Takagi-Sugeno-Kang (DG-TSK) fuzzy model to simultaneously conduct feature selection and rule extraction in one training phase. Firstly, a novel gate function is introduced, which is embedded into the system. We call this function as M-gate since it is shaped like an M. Secondly, a new type of fuzzy rule base (FRB) is designed by transforming each data point into one fuzzy rule, which is named point-based FRB (P-FRB). Based on the M-gate and P-FRB, DG-TSK model is developed, in which M-gates are embedded in the antecedents for feature selection and consequents for rule extraction. Note that these two procedures are performed at the same time. We test DG-TSK on 18 classification datasets and compare it with other similar methods. The experiment results demonstrate that DG-TSK is effective.

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Keywords: Takagi-Sugeno-Kang (TSK) model; Gate function; Feature selection; Rule extraction; Fuzzy system

1. Introduction

In recent years, feature selection has been attracting the interests of many scholars [1–4]. The number and quality of the features significantly affect the performance of the model. Good features have a positive impact on the given

☆ This work was supported in part by the National Key R&D Program of China under Grant 2019YFA0708700; in part by the National Natural Science Foundation of China under Grant 62173345; and in part by the Fundamental Research Funds for the Central Universities under Grant 20CX05002A, 22CX03002A.

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<https://doi.org/10.1016/j.fss.2023.108627>

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task, while bad features negatively influence the model's effectiveness. Basically, there are four kinds of features [5]: essential features, derogatory or bad features, indifferent features, and redundant features. Essential features are important for solving the given problem, while derogatory or bad features hinder problem-solving. Indifferent features have neither positive nor negative influences on the tasks, and as for redundant features, they are useful but not all of them are needed. The aim of the feature selection is to select the optimal subset of the features to improve the predictive ability of the method, reduce the storage burden, and enhance the interpretability of the model [6,7]. In other words, the essential features need to be selected. The bad and indifferent features should be discarded. The redundancy of the redundant features should be controlled at a low level [8].

In general, traditional feature selection methods can be classified into three groups, i.e., filter methods [9], wrapper methods [10,11] and embedded method [12,1], which are elaborated in [13]. In these three groups of feature selection methods, embedded method is more efficient than the other two ones as it selects features along with the system identification [14]. In [7,2], an embedded feature selection method was proposed in the frame of the neural networks. The authors add the $L_{2,1}$ regularization term to the loss function for a conventional multilayer perceptron (MLP) with one hidden layer. After the training, the input features with small weights are regarded as unimportant features and removed. By adding an L_1 norm penalty to the ordinary least squares, the authors in [15] introduced a novel interacted Lasso regression model to select features and identify high-order feature interactions. Xiang et al. [16] also combined the least squares regression with $L_{2,1}$ norm to achieve the feature selection procedure. In [17], the authors extended the traditional expectation-maximization (EM) algorithm, which was applied to estimate Gaussian mixture model (GMM) parameters, by adding a feature selection step. As a result, a new embedded approach of feature selection called expectation selection maximization (ESM) is developed.

Takagi-Sugeno-Kang (TSK) fuzzy system [18,19] is an effective technique to address nonlinear problems, such as classification, regression, and function approximation problems [20–23]. There have been some research achievements in embedded feature selection using TSK fuzzy models. Pal and Saha [24] proposed an integrated learning mechanism that can discriminate the bad features in the system identification for TSK fuzzy system. In this method, the feature modulators, a function to eliminate the effects of the bad features, are used both on antecedents and consequents to select features. Note that in the follow-up studies, the feature modulator is named as gate function since it acts like a gate [14]. In [25], the authors also used gate function on the fuzzy rules for feature selection, in which structure identification and parameter learning are integrated into a Bayesian model. In the above methods, the features with low gate values are going to be removed. After some features are removed, some fuzzy rules have the same antecedent. It is necessary to retain one of them and delete the rest of them. Therefore, the fuzzy rules are implicitly learned and selected. These methods are effective since the model's performance is considered and the computation cost is satisfactory. From the point of view of rule extraction, there is no measurement for each rule to compare which one is more important. The retained rules may not be optimal and the number of the rules can be reduced further.

The core and key point of a fuzzy system is its fuzzy rules including how many of them and what they are, which determines the clarity and predictive accuracy of the model [26]. Hence, it is pretty meaningful for TSK fuzzy model to design the effective rule extraction method to obtain high-quality rules. Abe and Lan [27] extracted fuzzy rules directly from the input data by recursively resolving overlaps between two classes. This extraction method is easy but the inference procedure is complicated. In [28], Setnes combined orthogonal least squares method and fuzzy clustering to extract fuzzy rules from data, in which the orthogonal least squares approach based reduction method was used to remove less important clusters. Ishibuchi and Yamamoto [29] used a multi-objective genetic local search algorithm to select the desirable fuzzy rules in the candidate ones generated by confidence and support metrics. In addition, Ishibuchi investigated some other rule extraction methods based on genetic algorithm [30–32]. By considering each support vector machine (SVM) as a fuzzy rule-based model, Castro et al. [33] extracted two fuzzy rules from each SVM to construct the fuzzy system, where every rule only depends on the set of support vectors. In addition, regularization term and Lasso method are also effective techniques to reduce the number of fuzzy rules. The authors in [34] applied the sparse coding for rule reduction, where L_1 norm was embedded in the system identification procedure. In [35], the antecedents of the fuzzy rules were constructed by the proposed fuzzy ordered c-means (FOCM) clustering method. As the corresponding fuzzy rule base (FRB) was redundant, the authors used the least angle regression (LAR) algorithm to obtain the consequents and select the important rules. In the very recent work [13], the authors proposed an integrated approach, called FSRE-AdaTSK, for feature selection and rule extraction, where adaptive softmin based TSK model is used. In FSRE-AdaTSK, the feature selection and rule extraction procedures are successively conducted using gate function in two separate training phases. After feature

selection, the FRB is extended to a big one, enhanced FRB (En-FRB), containing more rules for rule extraction. The whole training process is complex. It is necessary to investigate how to perform the feature selection and rule extraction procedures in the same FRB and in one training phase.

We note that, in this paper, rule extraction is considered as selecting a set of important rules from an FRB involving abundant fuzzy rules. According to [13], traditional fuzzy rule bases (FRBs) are divided into two categories, compactly combined fuzzy rule base (CoCo-FRB) and fully combined fuzzy rule base (FuCo-FRB). CoCo-FRB converts each cluster or fuzzy partition lying in the diagonal into one fuzzy rule, while FuCo-FRB considers all possible rules on the fuzzy partitions. In CoCo-FRB, the number of fuzzy rules is equal to the number of clusters or fuzzy partitions/sets defined on each feature, which means the FRB is too scarce for rule extraction unless amount of fuzzy sets are defined on each feature. In FuCo-FRB, the number of rules exponentially increases with the number of input features. For example, assume that the number of input features is 30 and the number of fuzzy sets/partitions defined on each feature is 3. Then the number of fuzzy rules generated in FuCo-FRB is 3^{30} . As a result, it is impractical for FuCo-FRB to be used in rule extraction approach. Besides, it is necessary to point out that the existing embedded methods for feature selection and rule extraction based on fuzzy model begin with a poor-quality FRB as some random initialization is included, whether the traditional CoCo-FRB, FuCo-FRB or En-FRB is used. How to begin feature selection and rule extraction with a high-quality FRB to improve the effectiveness of the algorithm is worthy of research.

The aim of this paper is to simultaneously realize the feature selection and rule extraction procedures in one training phase based on TSK fuzzy model. To distinguish which features and rules are essential, a new gate function is introduced which is called as M-gate result from that it is shaped like an M. In addition, a novel point-based FRB (P-FRB) is constructed to provide high-quality fuzzy model and abundant fuzzy rules at the beginning of feature selection and rule extraction. Based on M-gate and P-FRB, a TSK fuzzy model with double groups of gates (DG-TSK) is derived which is able to simultaneously select features and extract rules during training. The main contributions and advantages of this work are summarized as below.

- A new gate function called M-gate is introduced to pick out essential features and fuzzy rules. Different from the existing ones, M-gate is shaped like an M and achieves the maximum value of 1 at two points. On the other hand, M-gate is used in the same way as the traditional methods and, like some gate functions, it is an even function whose values are between 0 and 1.
- We propose a novel point-based FRB called P-FRB to guarantee the TSK fuzzy system has good performance and plenty of rules before training. Specifically, each data point is used to initialize an antecedent and the corresponding label encoded by one-hot encoding is regarded as the consequent. Without training, P-FRB based TSK fuzzy model has satisfactory ability for the given task. On the other hand, it includes abundant fuzzy rules as the number of rules is equal to the number of data point contained in the dataset.
- Using double groups of M-gates in the P-FRB based TSK fuzzy system, we develop so-called DG-TSK model. In the P-FRB based TSK fuzzy system, a group of proposed M-gates are embedded in the antecedents to select features. At the same time, another group of M-gates are embedded in the consequents to extract rules. These two groups of gates are optimized along with other system parameters in one training phase so as that the feature selection and rule extraction are simultaneously done. Since the feature space is reduced and FRB is simplified, the trained DG-TSK system has excellent interpretability in solving tasks.

The rest of this article is organized as follows. Traditional TSK fuzzy system and gate function based feature selection and rule extraction approaches are reviewed in the next section. Section 3 introduces the proposed M-gate, P-FRB and how to design DG-TSK. The effectiveness of the developed fuzzy model is demonstrated through the numerical simulations in Section 4. In the final section, we draw the conclusion and discuss the possible future research directions.

2. Preliminaries

In this section, the traditional TSK fuzzy system is first reviewed. Then, we introduce some gate functions and the existing approaches of feature selection and rule extraction based on them.

2.1. Traditional TSK fuzzy system

Consider a classification problem involving C classes. Let D be the number of features of the input data. $\mathbf{x} = (x_1, x_2, \dots, x_D) \in \mathbb{R}^D$ represents an instance or a data point. The number of fuzzy sets defined on each feature, x_d ($d = 1, 2, \dots, D$), is denoted by S . Here CoCo-FRB [13] is used to construct a TSK fuzzy model, in which the number of rules marked by R is equal to S . In general, for the first-order TSK fuzzy system with C -dimensional output, the r th ($r = 1, 2, \dots, R$) fuzzy rule is

$$\begin{aligned} \text{Rule}_r : & \text{IF } x_1 \text{ is } A_{r,1} \text{ and } \dots \text{ and } x_D \text{ is } A_{r,D}, \\ \text{THEN } & y_r^1(\mathbf{x}) = p_{r,0}^1 + \sum_{d=1}^D p_{r,d}^1 x_d, \dots, y_r^C(\mathbf{x}) = p_{r,0}^C + \sum_{d=1}^D p_{r,d}^C x_d, \end{aligned} \quad (1)$$

where IF clause (IF-part) is antecedent and THEN clause (THEN-part) is consequent. $A_{r,d}$ ($r = 1, 2, \dots, R$; $d = 1, 2, \dots, D$) is the r th fuzzy set defined on the d th feature, which is used in the r th rule. $y_r^c(\mathbf{x})$ ($c = 1, 2, \dots, C$) means the output of the r th rule associated with the c th class computed from a given instance $\mathbf{x} \in \mathbb{R}^D$. $p_{r,d}^c$ represents the consequent parameter of the r th rule associated with the d th feature for the c th class.

For the fuzzy set $A_{r,d}$, we use Gaussian membership function (MF) to model it. The membership values are calculated by

$$\mu_{r,d}(\mathbf{x}) = e^{-\frac{(x_d - m_{r,d})^2}{2\sigma_{r,d}^2}}, \quad (2)$$

where $m_{r,d}$ and $\sigma_{r,d}$ represent the center and spread of the r th Gaussian MF defined on the d th input variable, respectively. They are antecedent parameters that are going to be optimized. Note that, although in equation (2) the argument of $\mu_{r,d}$ is shown as $\mathbf{x}$, the function only uses the d th component of $\mathbf{x}$, x_d . This article is not dedicated to solving numeric underflow problem described in [13]. Hence, the firing strength of the r th fuzzy rule is defined through the commonly used product T-norm, i.e.,

$$f_r(\mathbf{x}) = \prod_{d=1}^D \mu_{r,d}(\mathbf{x}). \quad (3)$$

As shown in (1),

$$y_r^c(\mathbf{x}) = p_{r,0}^c + \sum_{d=1}^D p_{r,d}^c x_d. \quad (4)$$

The c th ($c = 1, 2, \dots, C$) component of the system output on $\mathbf{x}$ is computed by

$$y^c(\mathbf{x}) = \frac{\sum_{r=1}^R f_r(\mathbf{x}) y_r^c(\mathbf{x})}{\sum_{i=1}^R f_i(\mathbf{x})}. \quad (5)$$

The normalized firing strength of the r th rule on $\mathbf{x}$ is defined as

$$\bar{f}_r(\mathbf{x}) = \frac{f_r(\mathbf{x})}{\sum_{i=1}^R f_i(\mathbf{x})}. \quad (6)$$

Then (5) can be rewritten as

$$y^c(\mathbf{x}) = \sum_{r=1}^R \bar{f}_r(\mathbf{x}) y_r^c(\mathbf{x}). \quad (7)$$

This is the inference process of the traditional TSK fuzzy model with the product T-norm to obtain the system outputs, whose neural network structure is shown in Fig. 1. The first layer is the input layer. The second layer is the fuzzification layer, where the output of each node is computed by (2). In the third layer, D membership values are used together to compute a firing strength by a T-norm, such as the product T-norm, for each rule. Hence, this layer

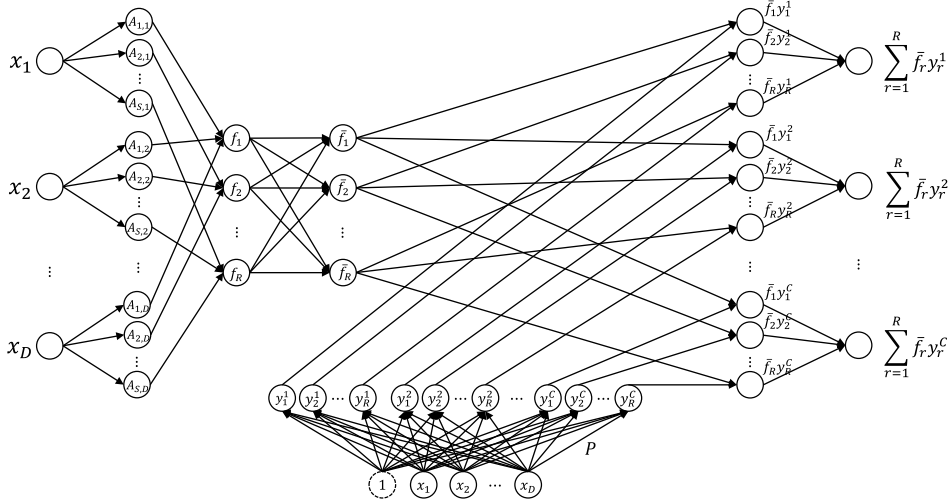


Fig. 1. The neural network structure of the first-order TSK fuzzy system.

is called rule layer. The firing strengths are normalized though (6) in the fourth layer. The rule outputs are computed by (4), which is shown as two fully connected layers in the lower part. Defuzzification process, (7), is realized by the last two layers.

The gradient based method is used to optimize the model. Here, we provide the gradient formulas of the system parameters with following the typical mean square error function

$$E = \frac{1}{2N} \sum_{n=1}^N \sum_{c=1}^C (y^c(\mathbf{x}_n) - y_c(\mathbf{x}_n))^2, \quad (8)$$

where N is the number of training samples, $y^c(\mathbf{x}_n)$ and $y_c(\mathbf{x}_n)$ respectively correspond to the c th component of the system output and the true label vector for the n th input instance, $\mathbf{x}_n$ ($n = 1, 2, \dots, N$). The gradients of the antecedent and consequent parameters are

$$\frac{\partial E}{\partial m_{r,d}} = \frac{1}{N} \sum_{n=1}^N \left[\bar{f}_r(\mathbf{x}_n) \frac{x_{n,d} - m_{r,d}}{\sigma_{r,d}^2} \sum_{c=1}^C [(y^c(\mathbf{x}_n) - y_c(\mathbf{x}_n))(y_r^c(\mathbf{x}_n) - y^c(\mathbf{x}_n))] \right], \quad (9)$$

$$\frac{\partial E}{\partial \sigma_{r,d}} = \frac{1}{N} \sum_{n=1}^N \left[\bar{f}_r(\mathbf{x}_n) \frac{(x_{n,d} - m_{r,d})^2}{\sigma_{r,d}^3} \sum_{c=1}^C [(y^c(\mathbf{x}_n) - y_c(\mathbf{x}_n))(y_r^c(\mathbf{x}_n) - y^c(\mathbf{x}_n))] \right], \quad (10)$$

and

$$\frac{\partial E}{\partial p_{r,d}^c} = \frac{1}{N} \sum_{n=1}^N [(y^c(\mathbf{x}_n) - y_c(\mathbf{x}_n)) \bar{f}_r(\mathbf{x}_n) x_{n,d}], \quad (11)$$

respectively, where $x_{n,d}$ is the d th component of $\mathbf{x}_n$. The parameters are updated in the k th iteration by

$$\omega^{(k+1)} = \omega^{(k)} - \eta \frac{\partial E}{\partial \omega^{(k)}}, \quad (12)$$

where ω represents the general parameter, $\eta > 0$ is the learning rate.

2.2. The gate function based feature selection and rule extraction approaches

For a T-norm, T , it always satisfies the following properties for all $v_1, v_2, v_3 \in [0, 1]$,

- 1) Commutativity: $T(v_1, v_2) = T(v_2, v_1)$,

- 2) Associativity: $T(v_1, T(v_2, v_3)) = T(T(v_1, v_2), v_3)$,
- 3) Monotonicity: $T(v_1, v_3) \leq T(v_2, v_3)$ if $v_1 < v_2$,
- 4) Neutrality: $T(v_1, 1) = T(1, v_1) = v_1$.

Obviously, product is a T-norm. The last property means 1 has no effect on the result of the T-norms. According to (3), if $\mu_{r,d'} = 1 \forall r = 1, 2, \dots, R$, the d' th feature makes no difference to the firing strength which should be removed. Some works [36,14,1] used the gate function to modulate the membership values in such way to remove the features whose modulated membership values are always 1. Specifically, the gate function, $M(\lambda)$, modulates the membership value, μ , as $\mu^{M(\lambda)}$, where λ is the gate parameter. In the feature selection approach, (2) is rewritten as

$$\mu_{r,d}(\mathbf{x}) = \left(e^{-\frac{(x_d - m_{r,d})^2}{2\sigma_{r,d}^2}} \right)^{M(\lambda_d)}, \quad (13)$$

where $M(\lambda_d)$ ($d = 1, 2, \dots, D$) is the gate function of the d th feature. In the training phase, λ_d s are updated by the gradient based method. When $M(\lambda_d) = 0$, the associated d th feature has no contribution to the firing strengths and the whole system. Hence it can be pruned.

As described in (1), the consequents are linear combinations of the input features in the first-order TSK model. The gate functions can be embedded in the consequents to select features. For all $r = 1, 2, \dots, R$ and $c = 1, 2, \dots, C$, $y_r^c(\mathbf{x})$ is modified by the following equation,

$$y_r^c(\mathbf{x}) = p_{r,0}^c + \sum_{d=1}^D M(\lambda_d) p_{r,d}^c x_d. \quad (14)$$

All the consequent parameters associated with the d th feature are multiplied by a same gate. Similarly, when $M(\lambda_d) = 0$, the d th feature has no influence on the consequents and the whole system, which can be removed. This is what the authors did for feature selection in [13].

In addition, the authors also used the gate function for rule extraction in [13]. For the r th fuzzy rule, assume that its gate function is denoted by $M(\theta_r)$ ($r = 1, 2, \dots, R$). The consequent is changed to

$$y_r^c(\mathbf{x}) = M(\theta_r) p_{r,0}^c + M(\theta_r) \sum_{d=1}^D p_{r,d}^c x_d. \quad (15)$$

All the consequent parameters in the r th rule are multiplied by a same gate. The rule gate parameters, θ_r s, are optimized along with other system parameters. $M(\theta_r) = 0$ means the r th rule contributes little to the system and needs to be eliminated.

The gate function is a function acting like a gate in the feature selection and rule extraction procedures. During the learning process, the gates should be closed for the bad features and unimportant rules and, on the other hand, the gates of the informative features and important rules should be opened. A key fact is that the opened gates should not affect the system, i.e., they can not help or hamper the model to predict the target outputs. Therefore, the gate function generally lies in the interval $[0, 1]$ or $[-1, 1]$. The commonly used ones are as below,

$$M(\lambda) = \frac{1}{1 + e^{-\lambda}}, \quad (16)$$

$$M(\lambda) = 1 - e^{-\lambda^2}, \quad (17)$$

$$M(\lambda) = e^{-\lambda^2}, \quad (18)$$

and

$$M(\lambda) = \lambda \sqrt{e^{1-\lambda^2}}, \quad (19)$$

where $\lambda \in \mathbb{R}$. The value of $M(\lambda)$ represents the opening of the gate, which indicates the importance of a feature or a rule. $M(\lambda) = 0$ means the gate is completely closed and the associated feature or rule is unimportant. On the other hand, $M(\lambda) = 1$ represents the gate is completely opened and the associated feature or rule is essential which is going

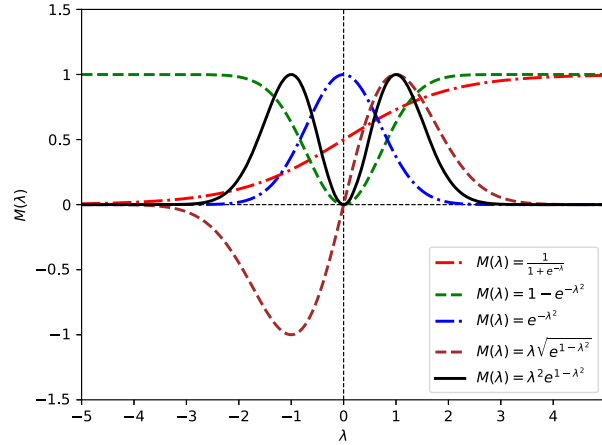


Fig. 2. Four existing gate functions and the proposed M-gate. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

to be selected or extracted. Note that for (13), (14) and (15), $M(\lambda) = 1$ does not affect the original membership values or rule outputs. When the last gate function, (19), is used in (14) or (15), both $M(\lambda) = 1$ and $M(\lambda) = -1$ mean the gate is open as the consequent parameters have no limit of positive or negative. But it can not be used on the antecedents for feature selection like $\mu^{M(\lambda)}$ as (19) has negative values which may result in the modulated membership values greater than 1. For example, if $\mu = 0.5$ and $M(\lambda) = -0.5$, the value of $\mu^{M(\lambda)}$ is 1.4142. This violates the fuzzy set theory and can not guarantee that the gate has no influence on the whole system if this gate is completely open.

3. Proposed method

This section presents the proposed M-gate, P-FRB and DG-TSK. The Python implementation of the DG-TSK model is available at <https://github.com/Eandon/DG-TSK>.

3.1. A new gate function: M-gate

At the beginning of the feature selection or rule extraction, all features or rules are regarded as bad features or useless rules. Hence, the gates should be initialized to low values. However, when the gate values are close to zero, the derivatives of the first three gate functions, (16)–(18), are also close to zero. This makes the learning very slow at the beginning. And unfortunately, as mentioned in the previous subsection, (19) can not directly be used on the antecedents for feature selection. In order to alleviate this problem, we propose the following gate function,

$$M(\lambda) = \lambda^2 e^{1-\lambda^2}, \quad (20)$$

where $\lambda \in \mathbb{R}$. This new gate function provides higher derivative than (16)–(18) when the gate values are the same low values. Fig. 2 depicts the plots of these five gate functions, in which the black solid line is our proposed gate function. Since (20) is shaped like an M, we name it as M-gate. The maximum value of M-gate locates in $\lambda = -1$ and 1 , which is different from the other four gate functions.

In Fig. 2, the plot of (16) is shown using the red dash-dotted line. The λ s are initialized at $-5.0 \pm \text{random noise}$ in $[0, 1]$ so that the gates of the features are closed at the beginning [37]. (17) and (18) are depicted by the green dashed line and blue dash-dotted line, respectively. The parameters, λ s, are initialized with 0.001 for the former [12]. While for the latter, λ s are initialized with $3.0 + \mathcal{N}(0, 0.2)$, where $\mathcal{N}(0, 0.2)$ indicates the Gaussian noise with mean zero and standard deviation 0.2 [14]. The brown dashed line depicts (19) which is an odd function. Its gate parameters are set to 0.01 before training in [13]. Following the previous works, to regard all the features and rules as unimportant ones at the very beginning, M-gates should be initialized to the low values (near zero). The gate parameter of the M-gate can be initialized to such as 0.01 or 0.1 to do that as it has bigger derivative at this region. In our method, we use M-gate for both feature selection and rule extraction.

3.2. A new fuzzy rule base: P-FRB

As explained in Section 1, neither CoCo-FRB nor FuCo-FRB is suitable for rule extraction. In addition, the initialized states of the existing FRBs are poor-quality which hampers the effect of the rule extraction. Aiming at conducting feature selection and rule extraction based on a high-quality FRB containing many of rules, we proposed a new FRB called P-FRB. This subsection introduces how to construct the P-FRB.

Suppose that the training set containing N data points along with their true labels is represented as

$$\mathbf{U} = \{(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N)\}, \quad (21)$$

where $\mathbf{x}_n$ and $\mathbf{y}_n$ ($n = 1, 2, \dots, N$) are the n th training sample with D features and the C -dimensional label vector encoded by one-hot encoding, respectively. For P-FRB, each data point contained in $\mathbf{U}$ is converted into a fuzzy rule, which means the number of the rules is N . More specifically, the antecedents include two parts of parameters, i.e., the centers and spreads, which are used in (2) to compute the membership values. The centers are initialized by

$$m_{r,d} = x_{r,d}, \quad (22)$$

where $x_{r,d}$ ($r = 1, 2, \dots, N$; $d = 1, 2, \dots, D$) is the value of the d th feature in the r th input sample. The spreads are initialized to the same value by

$$\sigma_{r,d} = \bar{\sigma} = \frac{1}{D} \sum_{d=1}^D \sqrt{\frac{\sum_{n=1}^N (x_{n,d} - \bar{x}_d)^2}{N-1}}, \quad (23)$$

where $\bar{x}_d = \frac{1}{N} \sum_{n=1}^N x_{n,d}$ is the mean of the d th feature of N training samples. $\bar{\sigma}$ is the mean of the standard deviations computed from the input domain according to the D features. The consequent parameters are initialized by

$$p_{r,d}^c = \begin{cases} y_{r,c}, & d = 0, \\ 0, & \text{otherwise,} \end{cases} \quad (24)$$

where $y_{r,c}$ is the c th component of the r th label vector, $\mathbf{y}_r$. In a word, we use the data points and their label vectors to initialize the antecedents and consequents for P-FRB, respectively.

For example, there are five data points with three features. Assume that they fall into two classes. The data points and their label vectors are expressed as below,

(0.2, 0.1, 0.4), (1, 0);
 (0.1, 0.1, 0.3), (1, 0);
 (0.3, 0.2, 0.3), (1, 0);
 (0.2, 0.4, 0.1), (0, 1);
 (0.1, 0.5, 0.3), (0, 1).

Based on the five data points, $\bar{\sigma} = 0.125$ can be obtained by (23). Replacing $A_{r,d}$ in the (1) with $e^{-\frac{(x_d - m_{r,d})^2}{2\sigma_{r,d}^2}}$, P-FRB has the following five rules,

IF x_1 is $e^{-\frac{(x_1 - 0.2)^2}{2 \times 0.125^2}}$ and x_2 is $e^{-\frac{(x_2 - 0.1)^2}{2 \times 0.125^2}}$ and x_3 is $e^{-\frac{(x_3 - 0.4)^2}{2 \times 0.125^2}}$, THEN $y_1^1(\mathbf{x}) = 1$, $y_1^2(\mathbf{x}) = 0$;
 IF x_1 is $e^{-\frac{(x_1 - 0.1)^2}{2 \times 0.125^2}}$ and x_2 is $e^{-\frac{(x_2 - 0.1)^2}{2 \times 0.125^2}}$ and x_3 is $e^{-\frac{(x_3 - 0.3)^2}{2 \times 0.125^2}}$, THEN $y_1^1(\mathbf{x}) = 1$, $y_1^2(\mathbf{x}) = 0$;
 IF x_1 is $e^{-\frac{(x_1 - 0.3)^2}{2 \times 0.125^2}}$ and x_2 is $e^{-\frac{(x_2 - 0.2)^2}{2 \times 0.125^2}}$ and x_3 is $e^{-\frac{(x_3 - 0.3)^2}{2 \times 0.125^2}}$, THEN $y_1^1(\mathbf{x}) = 1$, $y_1^2(\mathbf{x}) = 0$;
 IF x_1 is $e^{-\frac{(x_1 - 0.2)^2}{2 \times 0.125^2}}$ and x_2 is $e^{-\frac{(x_2 - 0.4)^2}{2 \times 0.125^2}}$ and x_3 is $e^{-\frac{(x_3 - 0.1)^2}{2 \times 0.125^2}}$, THEN $y_1^1(\mathbf{x}) = 0$, $y_1^2(\mathbf{x}) = 1$;
 IF x_1 is $e^{-\frac{(x_1 - 0.1)^2}{2 \times 0.125^2}}$ and x_2 is $e^{-\frac{(x_2 - 0.5)^2}{2 \times 0.125^2}}$ and x_3 is $e^{-\frac{(x_3 - 0.3)^2}{2 \times 0.125^2}}$, THEN $y_1^1(\mathbf{x}) = 0$, $y_1^2(\mathbf{x}) = 1$.

Now, we can compute the firing strengths by (3) and infer the system outputs for the five data points, which are

(0.9967, 0.0033);
 (0.9901, 0.0099);
 (0.9568, 0.0432);
 (0.0579, 0.9421);
 (0.0211, 0.9789).

According to the system outputs, each sample is assigned to the class closest to it. Moreover, it is obvious that the classification accuracy on these five data points is 100%. This case simply and clearly illustrates how to construct the P-FRB. On the other hand, it seems that, at the initial state (without training), P-FRB based TSK fuzzy system has good performance, which is going to be further investigated in Section 4.2.

We note here that for the binary classification problems, changing the form of the consequents, in which its sign would determine the class assignment, would reduce the number of the consequent parameters by half. More specifically, the consequent of the r th fuzzy rule is changed from “THEN $y_r^1(\mathbf{x}) = p_{r,0}^1 + \sum_{d=1}^D p_{r,d}^1 x_d$, $y_r^2(\mathbf{x}) = p_{r,0}^2 + \sum_{d=1}^D p_{r,d}^2 x_d$ ” to “THEN $y_r(\mathbf{x}) = p_{r,0} + \sum_{d=1}^D p_{r,d} x_d$ ”. The following decision function is commonly used to determine whether the sample belongs to the positive or negative class [38,39]:

$$\hat{y}(\mathbf{x}) = \text{sign}(y(\mathbf{x})) = \begin{cases} +1, & y(\mathbf{x}) \geq 0, \\ -1, & y(\mathbf{x}) < 0, \end{cases} \quad (25)$$

where $\hat{y}(\mathbf{x})$ and $y(\mathbf{x})$ are the predicted label and system output for the input sample $\mathbf{x}$, respectively. The rule of this form is more concise than the rule described in (1). However, our intention is to give a general methodology for both binary and multi-class problems. So the general rule shown in (1) is adopted in this article.

3.3. Double groups of gates based TSK model: DG-TSK

As illustrated in the previous subsection, P-FRB has a good performance and contains as many rules as the samples. In fact, rule extraction is necessary for P-FRB as it consists of so many rules. This subsection introduces how to use M-gate to simultaneously conduct feature selection and rule extraction based on P-FRB.

We use D M-gates on the antecedents of all fuzzy rules to assign a gate for every feature. The membership values computed from the d th feature are modulated by the gate function, $M(\lambda_d)$, which is placed in the exponential position of $\mu_{r,d}$. At the same time, R M-gates are embedded in the consequents for each rule. The r th M-gate, $M(\theta_r)$, is multiplied by all the consequent parameters in the r th rule. The usages of $M(\lambda_d)$ and $M(\theta_r)$ for feature selection and rule extraction have been shown in (13) and (15), respectively. The gradients of the error function with respect to λ_d and θ_r are provided in (26) and (27).

$$\frac{\partial E}{\partial \lambda_d} = M(\lambda_d) \left(\lambda_d - \frac{1}{\lambda_d} \right) \frac{1}{N} \sum_{n=1}^N \left[\sum_{r=1}^R \left[\bar{f}_r(\mathbf{x}_n) \frac{(x_{n,d} - m_{r,d})^2}{\sigma_{r,d}^2} \sum_{c=1}^C [(y^c(\mathbf{x}_n) - y_c(\mathbf{x}_n))(y_r^c(\mathbf{x}_n) - y^c(\mathbf{x}_n))] \right] \right]. \quad (26)$$

$$\frac{\partial E}{\partial \theta_r} = 2 \left(\frac{1}{\theta_r} - \theta_r \right) \frac{1}{N} \sum_{n=1}^N \left[\bar{f}_r(\mathbf{x}_n) \sum_{c=1}^C [(y^c(\mathbf{x}_n) - y_c(\mathbf{x}_n)) y_r^c(\mathbf{x}_n)] \right]. \quad (27)$$

Fig. 3(a) and (b) show which membership values and consequent parameters are tuned by the gate functions, respectively. The gate parameters, λ_d s and θ_r s, are simultaneously optimized along with the consequent parameters by the gradient based method. Note that the antecedent parameters are not included. As explained in [24], the tuning of the system parameters simultaneously with the gate parameters is not recommended. The reason why we simultaneously update the gate parameters and consequent parameters is that the tuning of the later can enhance the quality of the extracted rules in the numerical experiments.

After the training for feature selection and rule extraction, the features and rules are selected and extracted, respectively, according to the values of the M-gates. If the values of the feature gates are greater than the given threshold, τ_λ ,

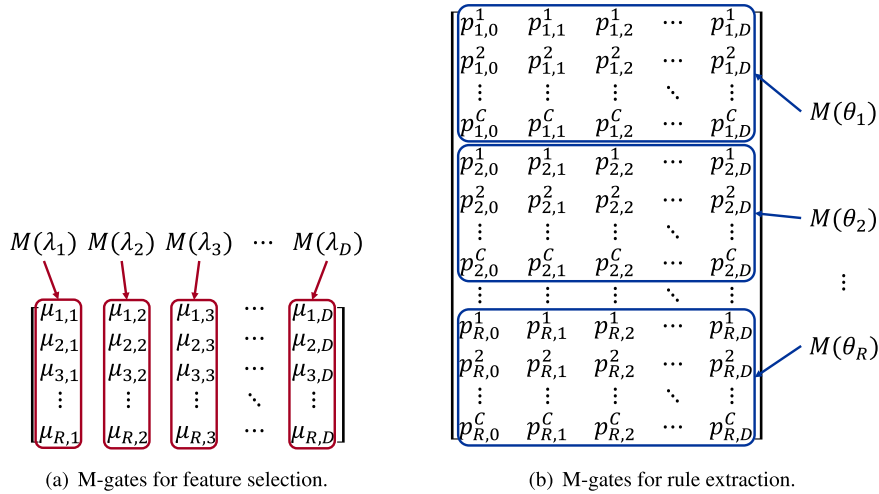


Fig. 3. The usage of gate functions for feature selection and rule extraction.

the corresponding features are selected, while others are removed. Similarly, the rules with the gate values bigger than the given threshold, τ_θ , are retained and others are pruned. The thresholds for feature selection and rule extraction are computed by

$$\tau_\lambda = \max_d \{M(\lambda_d)\} - \zeta_\lambda \left[\max_d \{M(\lambda_d)\} - \min_d \{M(\lambda_d)\} \right], \quad (28)$$

and

$$\tau_\theta = \max_r \{M(\theta_r)\} - \zeta_\theta \left[\max_r \{M(\theta_r)\} - \min_r \{M(\theta_r)\} \right], \quad (29)$$

respectively, where $\zeta_\lambda \in [0, 1]$ and $\zeta_\theta \in [0, 1]$ are the threshold coefficients which defined by the users. If a bigger ζ_λ is used, then more features will be selected. Similarly, the bigger ζ_θ is, the more rules are extracted. Finally, based on the selected features and extracted rules, the system is trained again for fine tuning, where both the antecedents and consequents are optimized. Before the training for fine tuning, all the consequent parameters are set to zero in our experiments. In order to describe the proposed method more clearly, the algorithm flow chart of DG-TSK is presented in Algorithm 1.

Even though the proposed DG-TSK is similar to the previous work [13], FSRE-AdaTSK, as they all work for feature selection and rule extraction, there are three main differences between them. 1) DG-TSK is a two-phase approach while FSRE-AdaTSK contains three training phases, which is the biggest difference. In FSRE-AdaTSK, the size of the En-FRB, proposed in [13] for rule extraction, is linearly dependent on the feature dimension, hence the rule extraction is performed after feature selection. In DG-TSK, these two procedures are performed based on the same FRB, i.e., P-FRB, which results in the DG-TSK has one less training phase. 2) The embedding strategies in DG-TSK and FSRE-AdaTSK are different. For FSRE-AdaTSK, the gate functions are embedded into the consequents both in feature selection and rule extraction procedures. In DG-TSK, the feature gates are embedded into the antecedents and, at the same time, the rule gates are embedded into the consequents. 3) A novel gate function called M-gate is used in DG-TSK, which is different from that, (19), in FSRE-AdaTSK. Note that (19) is not suitable for use on antecedent to select features as it has negative values. Hence, M-gate is used both on antecedents and consequents in DG-TSK.

4. Simulation results

To demonstrate the effectiveness of the proposed M-gate, P-FRB and DG-TSK, they are tested on 18 classification datasets in the experiments. The information of these datasets, including the number of features (#Features), the number of classes (#Classes) and the size of datasets, is summarized in Table 1.

As presented in Table 1, some datasets, like Page-blocks and Texture, have thousands of data points. Even after dividing the training and test set, plenty of samples are included in the training set, which means the initialized P-FRB

Algorithm 1 The algorithm of DG-TSK model.

Input: Training samples along with their class labels, $\mathbf{x}_n$ and $\mathbf{y}_n$, $n = 1, 2, \dots, N$; The learning rate, η ; The maximum number of training iterations for the feature selection and rule extraction phase and fine tuning phase, K ; The threshold coefficients, ζ_λ and ζ_θ .

Output: The final system and gate parameters.

- 1: Step 1. Initialize P-FRB through (22), (23) and (24). Calculate the forward propagation process by (13), (3), (6), (15) and (7), and compute the system error by (8). Then, compute the gradients of the error with respect to the consequent parameters ($p_{r,d}^c$ s) and gate parameters (λ_d s and θ_r s) by (11), (26) and (27), and update them K iterations by (12).
- 2: Step 2. Compute the thresholds, τ_λ and τ_θ , by (28) and (29), respectively. Select the features whose gate values are greater than the corresponding threshold, $M(\lambda_d) > \tau_\lambda$. Extract the rules whose gate values are greater than the corresponding threshold, $M(\theta_r) > \tau_\theta$.
- 3: Step 3. Based on the reduced feature space and simplified fuzzy rule base, set all the consequent parameters to zero and calculate the forward propagation process by (2), (3), (6), (4) and (7), and compute system error by (8). Then compute the gradients of the system error with respect to the centers ($m_{r,d}$ s), spreads ($\sigma_{r,d}$ s) and consequent parameters ($p_{r,d}^c$ s) by (9), (10) and (11), and update them K iterations by (12). This concludes design process of the rule base.

Table 1
Summary of the 18 classification datasets.

Dataset	#Features	#Classes	Dataset Size
Iris	4	3	150
Appendicitis	7	2	106
Pima	8	2	768
Yeast	8	10	1484
Glass	9	6	214
Page-blocks	10	5	5473
Wine	13	3	178
Heart	13	2	270
Vehicle	18	4	846
DRD	19	2	1151
IS	19	7	2310
Waveform	21	3	5000
Steel	27	7	1941
Wdbc	30	2	569
Texture	40	11	5500
Biodeg	41	2	1055
Spectfheart	44	2	267
Sonar	60	2	208

has so many rules. A great deal of rules cause a huge amount of computation which is not conducive to the feature selection and rule extraction. In the experiments, the number of rules created in P-FRB is set to a maximum of 300. If the number of the data points contained in the training set is more than 300, according to the stratified sampling strategy, 300 data points are randomly sampled to generate 300 fuzzy rules. For example, assume that the training set of Pima has 690 data points including 230 samples of class 1 and 460 samples of class 2. Hence $300 \times \frac{230}{690} = 100$ and $300 \times \frac{460}{690} = 200$ samples are randomly sampled from class 1 and class 2, respectively. Based on the sampled data points, the fuzzy rules are generated by (22), (23) and (24). For the datasets containing less than 300 data points, all of the samples are used to initialize P-FRB.

In addition, ten-fold cross-validation mechanism [14,7] is employed to minimize the effect of the initial settings, such as the division of the training and test set. It is repeated 10 times to report the average performance.

4.1. Comparisons of the gate functions

In this section, the proposed gate function, M-gate, is compared with the existing gate functions based on the DG scheme described in Section 3.3, where both feature selection and rule extraction are included. As mentioned before,

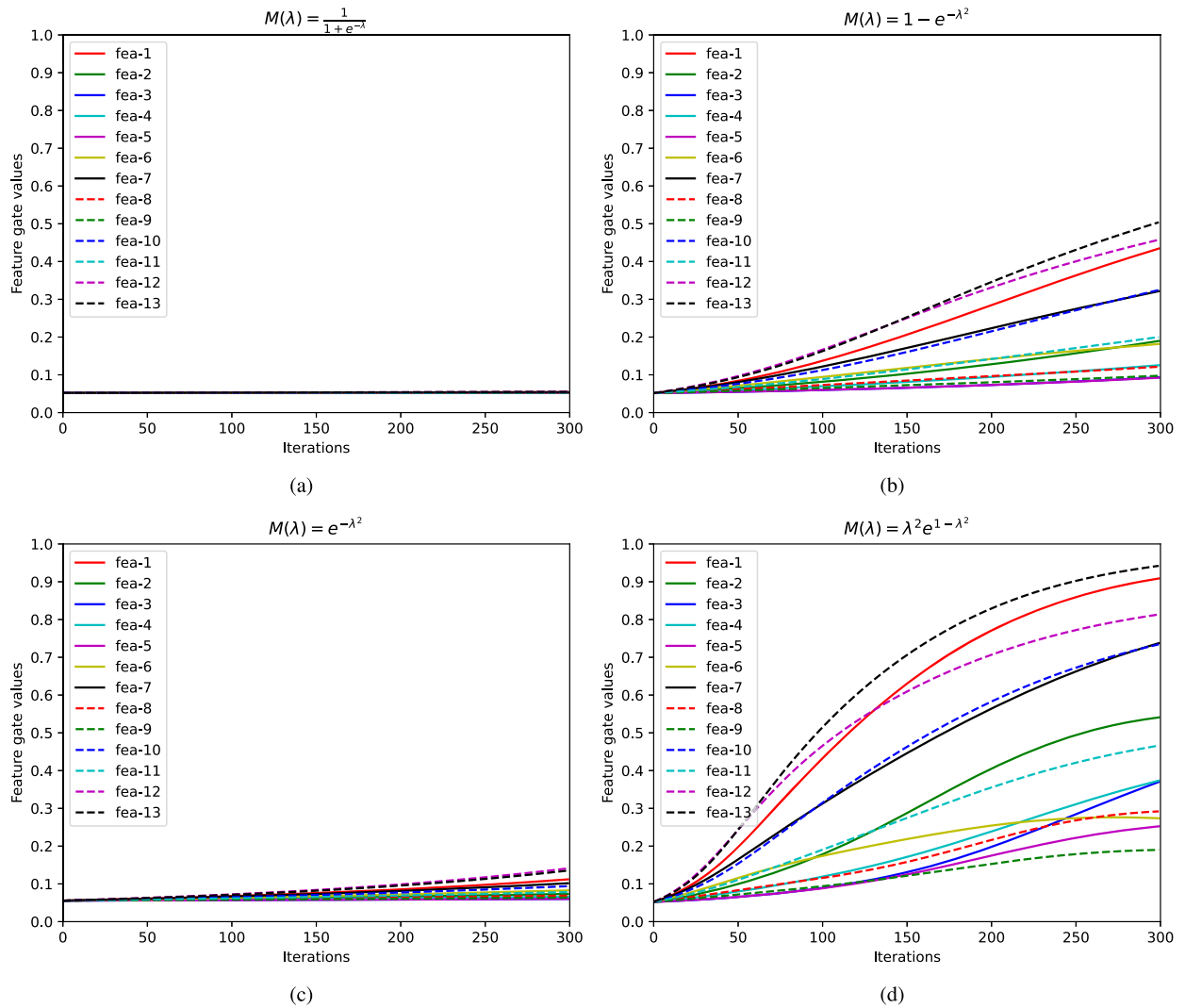


Fig. 4. The feature gate values of different gate functions on Wine dataset.

(19) is an odd function, which can not be embedded on the antecedents for feature selection. As a result, this odd gate function is not considered here. On Wine dataset with 13 features and 178 samples, all the samples are considered training samples. We construct a fuzzy system using the P-FRB introduced in Section 3.2, which means there are 178 fuzzy rules. Two groups of gates are simultaneously embedded into the system, where one group of feature gates are embedded into the antecedents for feature selection and the other group of rule gates are embedded into the consequents for rule extraction. All the gate values are initialized with the values around 0.05 to regard all the features and rules as unimportant ones at the beginning. The learning rate is set to 0.2. Full batch gradient descent method is used to optimize the system 300 iterations. The feature and rule gate values based on four different gate functions over 300 iterations are shown in Fig. 4 and 5, respectively.

From Fig. 4, we can see that the first and third gate function do not preferably distinguish which features are important. The second gate function gives the passable results, but they are not as good as those of M-gate. In Fig. 4(d), M-gate satisfactorily scatters each feature and shows which features are important. In addition, Fig. 5 illustrates the advantages of M-gate more clearly. When dealing with 178 rule gates, it seems that the three existing gate functions are not the good choices as they can not satisfactorily scatter the rules. M-gate can still tell which rules are more important. On more complex datasets with more features and samples, M-gate will have a more obvious advantage.

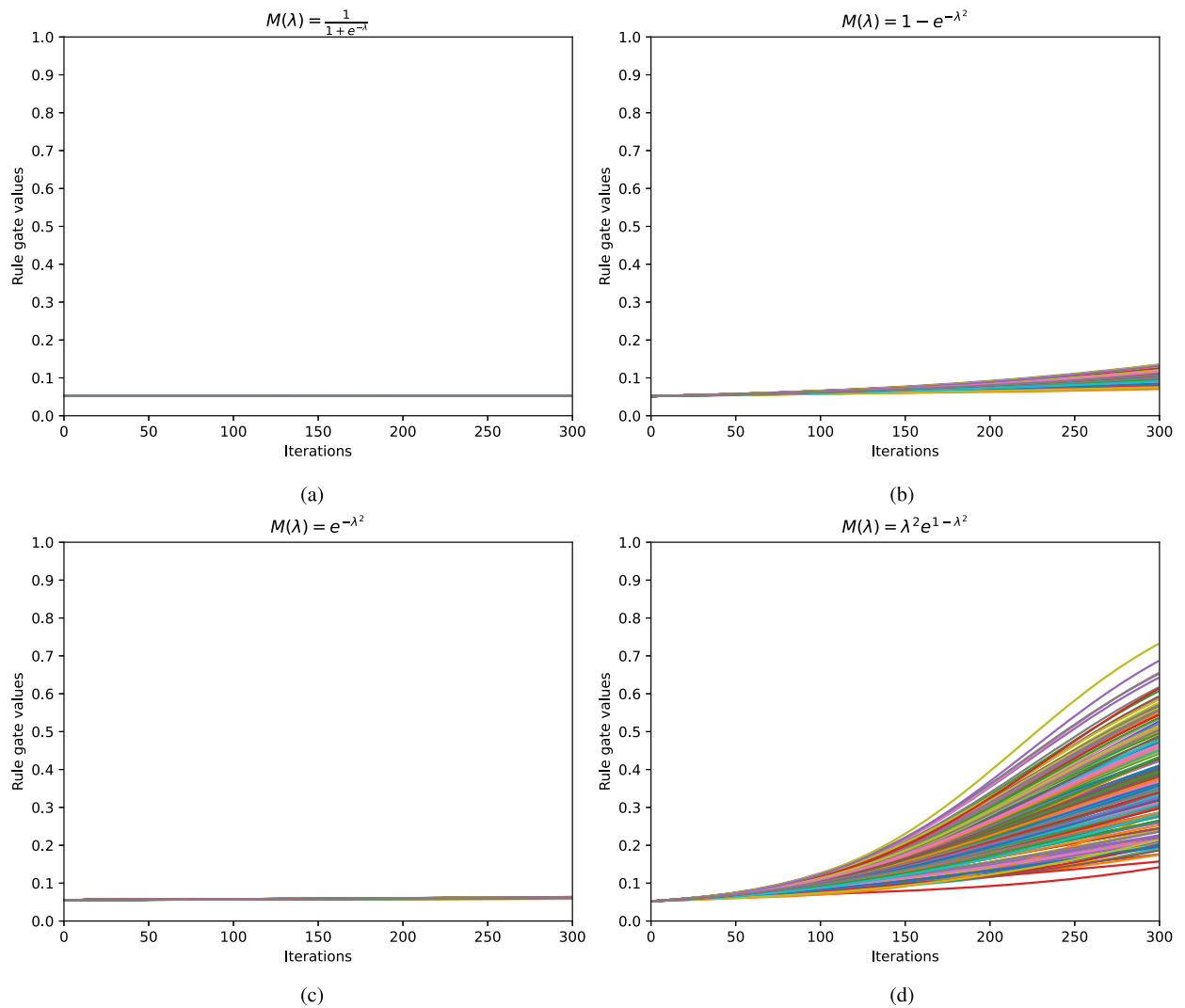


Fig. 5. The rule gate values of different gate functions on Wine dataset.

From Fig. 4(b) and 4(d), it can be seen that the proposed gate function can learn to distinguish between useful and poor features faster than the existing one with the same learning rate and iterations. A bigger gate value means the corresponding feature is more important. These two groups of experiments with different gate functions give the very similar results, i.e., the features 13, 1, 12, 7 and 10 are more important than others. This implicitly demonstrates the effectiveness of the proposed gate function in DG-TSK model.

4.2. The classification performance of P-FRB based TSK

To illustrate that the proposed P-FRB, at its initial state, can provide good performance for TSK fuzzy system, we test it on the 18 classification datasets listed in Table 1. Note that P-FRB based TSK is without training. In other words, P-FRB based TSK is used for test tasks as soon as it is initialized. We do not pursue the best performance for P-FRB based TSK model among the recently proposed classification methods. Two classic classifiers, including random forest (RF) and support vector machine (SVM), and the recent proposed adaptive softmin based TSK model (AdaTSK) are chosen as the baselines of the classification experiments. The classification results of these three classifier and two TSK models are reported in Table 2, where the best results are marked in bold.

Table 2

Classification accuracies of RF, SVM, AdaTSK, random FRB based TSK and P-FRB based TSK.

Dataset (#Features)	RF	SVM	AdaTSK	Random FRB based TSK	P-FRB based TSK
Iris (4)	0.9507	0.9587	0.9547	0.2800	0.9400
Appendicitis (7)	0.8698	0.8678	0.8707	0.4585	0.8671
Pima (8)	0.7680	0.7678	0.7615	0.5013	0.7354
Yeast (8)	0.6244	0.5958	0.4930	0.0906	0.5562
Glass (9)	0.7963	0.6532	0.5397	0.1756	0.6400
Page-blocks (10)	0.9747	0.9446	0.8979	0.2024	0.9298
Wine (13)	0.9831	0.9866	0.9865	0.3410	0.9663
Heart (13)	0.8267	0.8237	0.8459	0.5348	0.7941
Vehicle (18)	0.7515	0.7507	0.6215	0.2489	0.6752
DRD (19)	0.6801	0.6739	0.6061	0.5065	0.6257
IS (19)	0.9795	0.9386	0.7823	0.1389	0.8952
Waveform (21)	0.8536	0.8656	0.8500	0.3403	0.8067
Steel (27)	0.7859	0.7133	0.6027	0.1521	0.6525
Wdbc (30)	0.9603	0.9765	0.9427	0.5372	0.9638
Texture (40)	0.9788	0.9918	0.8357	0.0928	0.9178
Biodeg (41)	0.8756	0.8561	0.8034	0.5142	0.8287
Spectfheart (44)	0.8126	0.7880	0.7985	0.5645	0.7187
Sonar (60)	0.8249	0.8392	0.7507	0.4909	0.8548

Since the maximum number of the rules in P-FRB based TSK is set to 300, the number of the trees used in RF is set to 300 in the experiments. For SVM, Gaussian kernel is used. As for AdaTSK, following [13] three fuzzy rules are used on each dataset. The centers are initialized with the equidistant points located in the interval defined by the minimum and the maximum feature values in the training set. More specifically, the centers are given by

$$m_{r,d} = x_d^{\min} + (x_d^{\max} - x_d^{\min}) \frac{r-1}{R-1}, \quad (30)$$

where $r = 1, 2, \dots, R$, $d = 1, 2, \dots, D$, $x_d^{\min}$ and $x_d^{\max}$ represent the minimum and maximum value of the d th feature on the input domain, respectively. The MF used in [13] is a simplified Gaussian function, $\mu_{r,d}(\mathbf{x}) = e^{-(x_d - m_{r,d})^2}$, whose spreads are equivalent to be fixed $\sqrt{\frac{1}{2}}$ in (2). All the consequent parameters are initialized to zero. After initialization, AdaTSK are trained a given number of iterations to fit the training set. Three rules are not by no means the optimal choice to deal with the datasets mentioned in Table 1 as some of them, such as Yeast and Texture, involve more than three classes. But its result is also a reasonable baseline. In addition, P-FRB based TSK is compared with random FRB based TSK where the consequents of the latter are randomly initialized to the values evenly distributed in the interval $[0, 1]$. The antecedents of random FRB based TSK are the same as those of P-FRB based TSK.

From Table 2, it can be seen that P-FRB based TSK without training achieves satisfactory classification results. Such as on Iris, Appendicitis and Wdbc, the results are close to the best ones. Especially on Sonar dataset, it gives the best result in the five models. As shown in the last two columns in Table 2, P-FRB based TSK fuzzy system outperforms the system with randomly initialized FRB. It can be concluded that P-FRB is suitable to be used in DG-TSK for feature selection and rule extraction as it has good initial performance and abundant rules.

It is easy to imagine that the performance of P-FRB based TSK fuzzy system without training depends on the spreads in the antecedents. We also observe this phenomenon when constructing P-FRB in our experiments. For P-FRB described in Section 3.2, the spreads are initialized to the same value, $\bar{\sigma}$, by (23). To investigate the influence of $\bar{\sigma}$ on the P-FRB based TSK, it is set to the value of $\{0.1, 0.2, \dots, 1.0\}$ to compare the different classification results. In Fig. 6, the average test accuracies are plotted in solid line for each datasets. The horizontal dashed line is the

classification accuracy of P-FRB based TSK reported in Table 2, in which the spreads are initialized with the mean of the standard deviations. As can be seen from Fig. 6, the results represented by the dashed line are not always the best. The classification performance on some datasets decreases sharply with the increase of $\bar{\sigma}$, such as Glass, Wdbc, Texture and Vehicle. On some other datasets, the performance of P-FRB based TSK does not fluctuate greatly, such as Page-blocks, Heart and DRD. Different from others, the classification accuracies on Heart and Spectfheart increase with the value of $\bar{\sigma}$ increasing. Therefore, without training, the value of $\bar{\sigma}$ is a key point for P-FRB based TSK. It is permissible to set the spreads to different values, which is not further studied here. It is recommended to use the mean of the standard deviations or the small values like 0.1 to initialize $\bar{\sigma}$. Note that (23) is used in this paper.

4.3. The classification performance of DG-TSK

In this subsection, DG-TSK is tested for feature selection and rule extraction on the 18 datasets. The P-FRB is constructed as described in Subsection 3.2. Two groups of M-gates are embedded in the P-FRB based TSK through (13) and (15), respectively, where the embedding location is shown in Fig. 3. In the experiments, the gate parameters, λ_{ds} and θ_r s, are initialized with 0.1. By doing this, the gate values are 0.0269 at the beginning, then all the features and rules are regarded as bad ones.

For feature selection, we set the threshold coefficient, ζ_λ , to 0.5. This is a moderate value for ζ_λ to select features, which is also used in [13]. While for rule extraction, we use the strict strategy by setting the threshold coefficient, ζ_θ , to 0.01. There are hundreds of fuzzy rules generated in P-FRB, which means most of them are supposed to be pruned. Therefore, the strict strategy is used here. On the other hand, we set a lower bound on the number of extracted rules, which is the number of classes involved in the dataset, C . This is to say, when the number of extracted rules is smaller than the number of classes, C fuzzy rules with the biggest gate values are extracted. Extracting at least C rules is not necessary, but it will facilitate the learning of the classifier especially in the case of a small ζ_θ . In fact, the optimal values of ζ_λ and ζ_θ can be acquired through ten-fold cross-validation experiments or some evolutionary algorithms. Here, we verify the validity of the proposed method by setting them the fixed values without searching the optimal values.

DG-TSK is compared with FSCR-FRBS [1] and FSRE-AdaTSK [13], where FSCR-FRBS is a feature selection method with controlled redundancy using a fuzzy rule based system and FSRE-AdaTSK is the method mentioned in Section 1. The results of them are listed in Table 3. The average classification accuracy (%), average number of selected features and average number of extracted rules are reported, which are denoted by Acc, #F and #R, respectively, where the best classification accuracies are marked in bold. In the last row, the average results on the 18 datasets are provided. The ranks of the Acc, #F and #R of the three fuzzy models are listed in Table 4 for a clearer comparison. Since higher accuracy means the performance of the model is better, we rank them from the highest accuracy to the lowest accuracy. While for the number of selected features and extracted rules, fewer of them means better and more interpretable models. Hence, the fuzzy algorithms are ranked in ascending order according to #F and #R, respectively, where the highest ranks in average results among the three fuzzy algorithms are marked in bold.

From Table 3 and 4, it is easy to see that our proposed DG-TSK method outperforms its two counterparts on most of the datasets. Especially on Spectfheart dataset, DG-TSK achieves higher classification accuracy with fewer features and rules. The average accuracy of DG-TSK on 18 datasets is 81.42%, which is the best one. Its rank reported in Table 4 also demonstrates this. In the experiments, we do not pursue the fewest features and rules. The problem of controlling redundancy in the set of selected features is not considered in DG-TSK. Hence, the number of selected features of DG-TSK associated with different datasets are not always less than those of the other two methods. Even so, the results of feature selection are acceptable. In addition, comparing with FSCR-FRBS and FSRE-AdaTSK, DG-TSK extracts fewer fuzzy rules on all the datasets except Heart and obtains satisfactory results, which is shown in Table 4. On half of the datasets, the number of extracted rules is equal to the number of classes. While on the other half of the datasets, the number of extracted rules is greater than the number of classes. This means that the setting of a small value and the lower bound for ζ_θ is helpful.

Comparing the results of DG-TSK with the results reported in Table 2, DG-TSK does not outperform the RF and SVM on most of datasets. This is because the number of the features used in DG-TSK is less than that of RF and SVM. During the feature selection procedure, the features that is not essential but has some classification information are removed, which may reduce the predictive ability of the model. But we emphasize that our proposed DG-TSK has good interpretability as its rule base is concise. After feature selection and rule extraction, the rule length is shorted

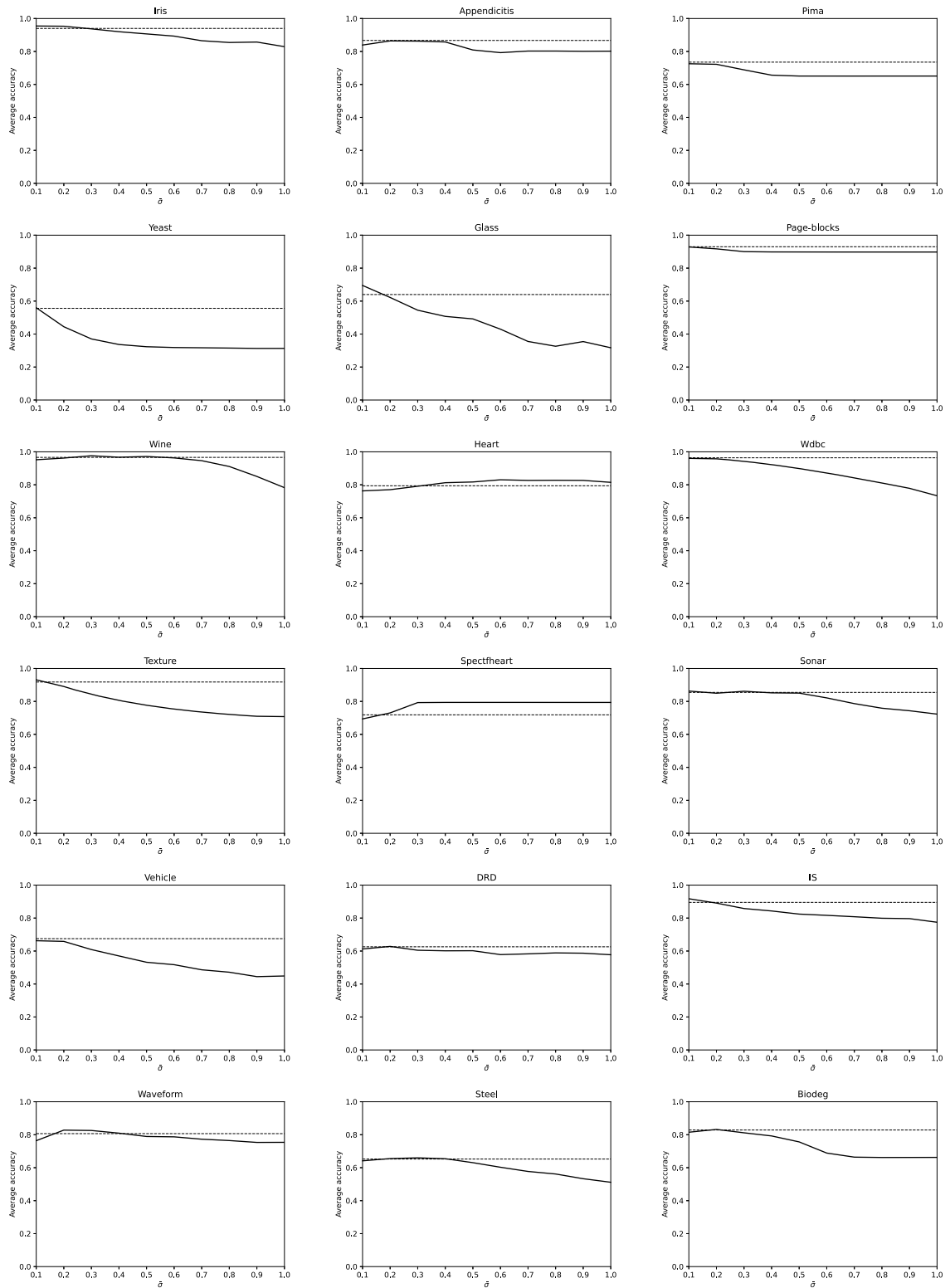


Fig. 6. The test accuracies of P-FRB based TSK (without training) with different spreads on the 18 datasets.

Table 3
The performance of DG-TSK on the 18 classification datasets.

Dataset (#Features)	FSCR-FRBS [1]			FSRE-AdaTSK [13]			DG-TSK (ours)		
	Acc	#F	#R	Acc	#F	#R	Acc	#F	#R
Iris (4)	96.0	1.0	5.4	96.5	2.1	6.1	96.8	2.3	3.1
Appendicitis (7)	84.6	3.8	3.4	86.3	3.7	3.1	86.6	5.3	2.1
Pima (8)	75.4	4.5	10.8	75.7	3.0	3.2	75.9	4.5	2.2
Yeast (8)	59.5	7.5	16.4	58.1	4.5	10.0	56.2	4.9	10.0
Glass (9)	65.2	7.1	9.6	65.5	4.1	6.0	63.1	8.0	6.0
Page-blocks (10)	94.2	5.5	6.0	93.4	2.2	5.6	91.5	2.8	5.0
Wine (13)	96.7	4.2	7.6	97.3	6.3	6.3	98.3	8.0	3.0
Heart (13)	80.4	7.3	12.7	82.5	7.4	2.9	85.0	4.0	3.4
Vehicle (18)	68.4	7.4	6.6	69.8	7.0	4.0	70.3	12.0	4.0
DRD (19)	65.7	4.5	4.9	62.8	3.5	2.4	67.4	8.7	2.0
IS (19)	90.5	9.0	17.4	90.3	9.8	7.0	91.7	14.2	7.0
Waveform (21)	77.5	12.2	9.0	85.3	15.0	8.2	86.4	15.8	3.0
Steel (27)	53.9	7.0	7.8	62.0	14.7	7.0	65.1	11.2	7.0
Wdbc (30)	94.6	3.0	7.7	95.4	6.2	4.9	96.2	5.0	2.2
Texture (40)	94.3	14.8	30.3	97.2	23.7	11.0	97.0	34.4	11.0
Biodeg (41)	75.9	3.5	3.6	80.5	8.7	2.3	80.8	6.2	2.1
Spectfheart (44)	79.7	3.3	4.0	79.2	3.4	5.1	79.9	2.9	2.1
Sonar (60)	76.7	8.3	7.4	73.4	10.5	2.1	77.4	12.7	2.0
Average	79.40	6.33	9.48	80.62	7.54	5.40	81.42	9.05	4.29

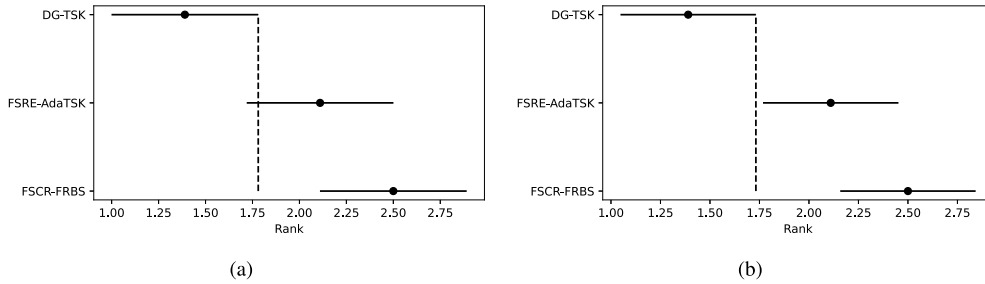
Table 4
The Ranks of the three fuzzy algorithms on the 18 classification datasets.

Dataset (#Features)	FSCR-FRBS [1]			FSRE-AdaTSK [13]			DG-TSK (ours)		
	Acc	#F	#R	Acc	#F	#R	Acc	#F	#R
Iris (4)	3	1	2	2	2	3	1	3	1
Appendicitis (7)	3	2	3	2	1	2	1	3	1
Pima (8)	3	2.5	3	2	1	2	1	2.5	1
Yeast (8)	1	3	3	2	1	1.5	3	2	1.5
Glass (9)	2	2	3	1	1	1.5	3	3	1.5
Page-blocks (10)	1	3	3	2	1	2	3	2	1
Wine (13)	3	1	3	2	2	2	1	3	1
Heart (13)	3	2	3	2	3	1	1	1	2
Vehicle (18)	3	2	3	2	1	1.5	1	3	1.5
DRD (19)	2	2	3	3	1	2	1	3	1
IS (19)	2	1	3	3	2	1.5	1	3	1.5
Waveform (21)	3	1	3	2	2	2	1	3	1
Steel (27)	3	1	3	2	3	1.5	1	2	1.5
Wdbc (30)	3	1	3	2	3	2	1	2	1
Texture (40)	3	1	3	1	2	1.5	2	3	1.5
Biodeg (41)	3	1	3	2	3	2	1	2	1
Spectfheart (44)	2	2	2	3	3	3	1	1	1
Sonar (60)	2	1	3	3	2	2	1	3	1
Average	2.50	1.64	2.89	2.11	1.89	1.89	1.39	2.47	1.22

Table 5

Ranks and p -value obtained by Friedman's test on the classification accuracies.

Method	Average rank	p-value
DG-TSK	1.39	
FSRE-AdaTSK	2.11	0.0033
FSCR-FRBS	2.50	

Fig. 7. Nemenyi's test. (a) $\alpha = 0.05$; (b) $\alpha = 0.1$.

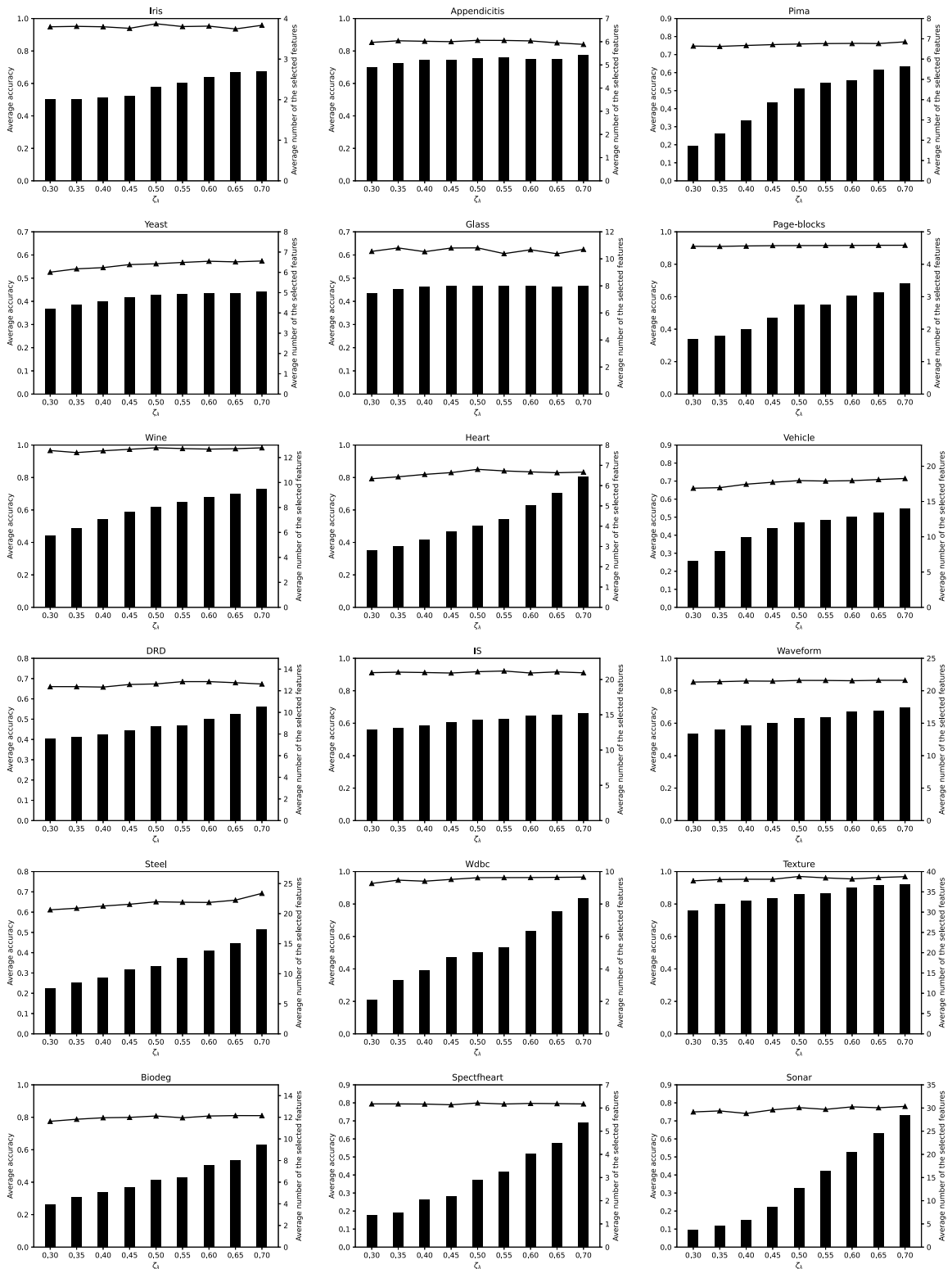
and the number of rules is reduced. For example, on Spectfheart dataset, only an average of 2.9 features and 2.1 rules are included in the final FRB of DG-TSK. It can be concluded that the proposed DG-TSK provides an improvement in the interpretability.

To further compare the performance of the three algorithms over multiple datasets, the nonparametric statistical tests are performed on the accuracies reported in Table 3. The Friedman's test [40] is conducted to check whether there are significant differences between these three groups of results. Then, Nemenyi's test [41] is carried out as a *post hoc* test to compare the methods. Table 5 presents the results of the Friedman's test. Since the p -value is less than the typical value of the level of confidence, i.e., $\alpha = 0.05$, the results claim that the significant differences exist among the three groups of results. Fig. 7 shows the critical difference (CD) in the Nemenyi's test for the three algorithms based on $\alpha = 0.05$ and 0.1. Note that the CD value is calculated by $q_\alpha \sqrt{\frac{K(K+1)}{6L}}$, where K and L represent the number of algorithms and datasets, respectively. With $K = 3$, q_α equals 2.344 and 2.052 when α is 0.05 and 0.1, respectively. It can be seen from Fig. 7 that the proposed DG-TSK method is significantly better than FSCR-FRBS. DG-TSK outperforms FSRE-AdaTSK when $\alpha = 0.1$ but is not significantly better than it when $\alpha = 0.05$. The results still support that the proposed DG-TSK is effective.

4.4. The sensitivity of DG-TSK to ζ_λ and ζ_θ

ζ_λ and ζ_θ determine how many features and rules are selected and extracted, respectively, which play an important role in DG-TSK. In the previous subsection, they are fixed at 0.5 and 0.01. Since $\zeta_\lambda = 0.5$ was used in [13], it is followed here. The lower bound of the number of the extracted rules is set to the number of classes, we use $\zeta_\theta = 0.01$ as strict strategy to remove most of the rules. The numerical experiments are designed in this subsection to investigate the sensitivity of DG-TSK to ζ_λ and ζ_θ .

In the sensitivity experiment associated with ζ_λ , it is set to the values of $\{0.3, 0.35, 0.4, \dots, 0.7\}$ while ζ_θ is fixed at 0.01. The average test accuracies and the average number of the selected features on each dataset are plotted, which are shown in Fig. 8. The horizontal axis is the values of ζ_λ . The left and right vertical axes are the average test accuracies and the average number of the selected features, respectively. As shown in Fig. 8, there is a marked increase for the number of the extracted rules as the value of ζ_λ gets bigger on most of datasets. But for the accuracy, it does not significantly fluctuate. Especially on Sonar dataset, even though the number of selected features increases from 3.70 to 28.36, the accuracy is not improved much. This phenomenon indicates that the useful features are selected by DG-TSK and the performance of our proposed method is not sensitive to ζ_λ .

Fig. 8. The sensitivity of DG-TSK to ζ_λ .

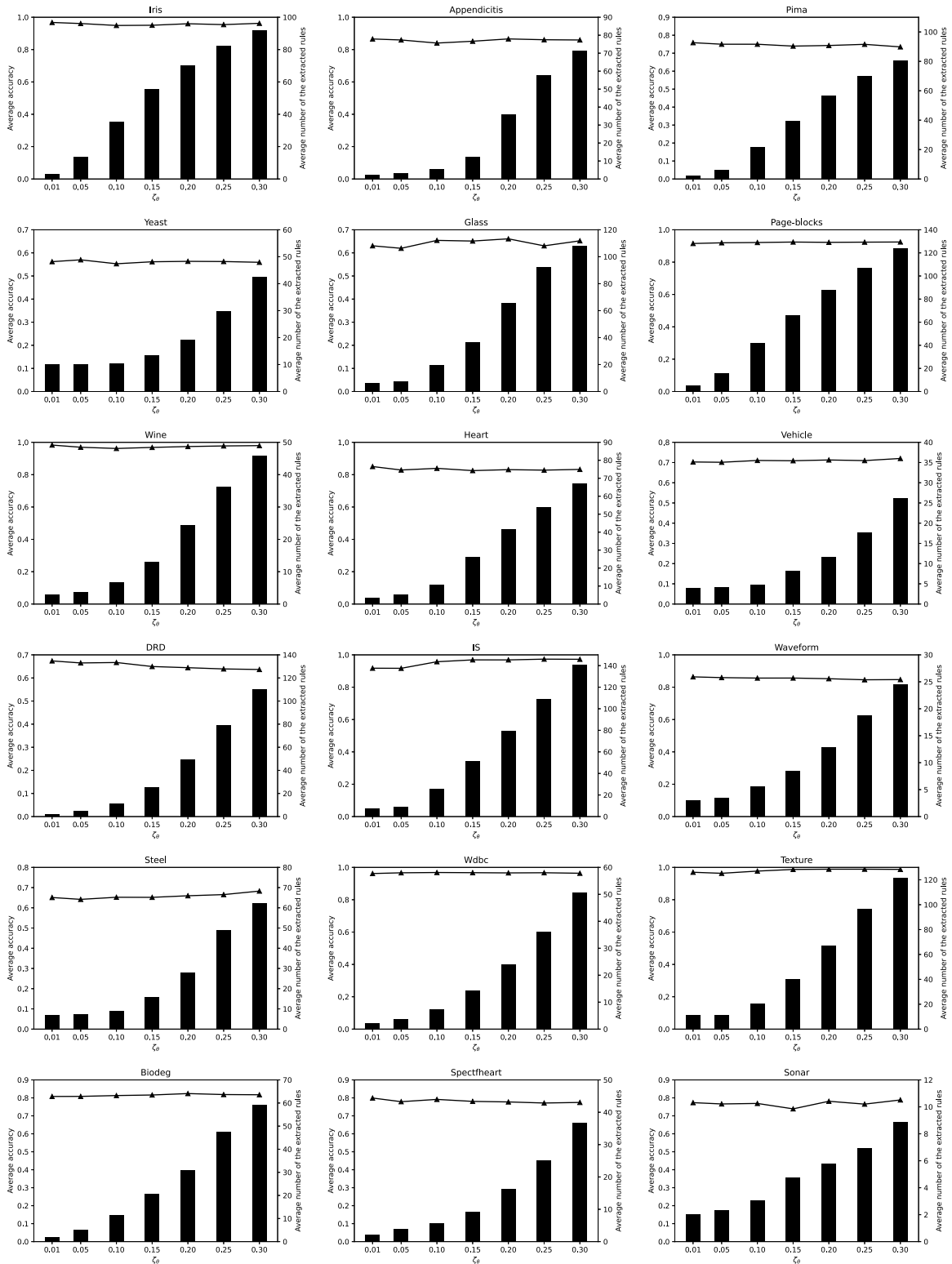
Fig. 9. The sensitivity of DG-TSK to ζ_θ .

Table 6
The computational complexity comparison.

Methods	Computational complexity	No. of phases
FSCR-FRBS	$O(D^2 + RD + D'^2 + RD')$	2
FSRE-AdaTSK	$O(CDS + CD'^2S + CD'R')$	3
DG-TSK	$O(CDR + CD'R')$	2

In the sensitivity experiment associated with ζ_θ , it is set to the values of $\{0.01, 0.05, 0.1, \dots, 0.3\}$. Similarly, when we investigate the sensitivity to ζ_θ , ζ_λ is fixed at 0.5. The results including accuracies and the number of extracted rules are plotted in Fig. 9. The reason we do not set larger values for ζ_θ to observe the sensitivity of DG-TSK is that the number of extracted rules becomes pretty large when ζ_θ is equal to 0.3. Such as on Glass, Page-blocks and Texture datasets, more than 100 rules are extracted in the case of $\zeta_\theta = 0.3$. In Fig. 9, the horizontal axis is the values of ζ_θ . The left and right vertical axes are the average test accuracies and the average number of the extracted rules, respectively. Comparing with the selected features, the extracted rules increase more markedly with ζ_θ increasing. We use the ideal labels to initialize the consequents, hence each rule contains useful knowledge which can be regard a good one. Therefore, all the rules should be extracted by M-gates. This explains why the number of extracted rules sharply increase with ζ_θ . However, it can be seen from Fig. 9 that there is no significant improvement in accuracy even if more rules are extracted. For example, on Page-blocks dataset, the number of extracted rules increases from 15.54 to 124.14, while the accuracy changes barely. ζ_θ does not need to be set to a large value in DG-TSK. Besides, the performance of DG-TSK is not sensitive to ζ_θ .

Nonetheless, it should be pointed out that the numbers of the selected features and the extracted rules are sensitive to the values of ζ_λ and ζ_θ , respectively. Especially for rule extraction approach, the number the extracted rules at $\zeta_\theta = 0.3$ is more than ten or even twenty times that at $\zeta_\theta = 0.01$. The threshold is computed based on ζ which is an artificially defined coefficient. To some extent, this hampers the efficiency of DG-TSK in selecting features and extracting rules. With proper ζ_λ and ζ_θ , DG-TSK is quite effective and efficient. This needs further study.

4.5. Computational complexity

In this subsection, we analyze the computational complexity of the proposed DG-TSK. For each instance, $R \times D$ membership values are computed by (13) and D M-gate values are calculated, where the computational complexity is $O(7RD + 5D)$, i.e., $O(DR)$. Then, R firing strengths and their normalized values are computed whose computational complexities are $O(D)$ and $O(R)$, respectively. As the R consequents with M-gates are calculated by (15), the computational complexity is $O(5R + (2D + 1)RC)$, that is to say $O(CDR)$. For the defuzzification process, the computational complexity is $O(CR)$. After the feature selection and rule extraction training, there is a fine-tuning phase. It is easy to derive that its computational complexity is $O(CD'R')$, where D' and R' are the number of the selected features and extracted rules, respectively. As a consequence, the computational complexity for the forward propagation of DG-TSK is $O(CDR + CD'R')$. Similarly, the computational complexity of the back-propagation for each instance is derived, which is also $O(CDR + CD'R')$. Thus, the overall complexity of the proposed DG-TSK model is $O(CDR + CD'R')$.

For comparison, Table 6 lists the computational complexities of the three fuzzy algorithms, where the last column is the number of the phases contained in them. The computational complexity of FSRE-AdaTSK is represented using S instead of R , which results from that R is not always equal to S . FSCR-FRBS has feature selection phase and fine tuning phase, which are two independent phases. While FSRE-AdaTSK contains three phases, i.e., feature selection, rule extraction and fine tuning phase. As for our proposed DG-TSK, it includes two phases, where the feature selection and rule extraction procedures are contained in the first phase and the second phase is the fine tuning phase.

From Table 6, it can be seen that the computational complexity of DG-TSK is not heavy comparing with that of the other two models. Even though, we have to confess that DG-TSK is time-consuming since the feature selection and rule extraction are simultaneously conducted based on all the features and abundant rules. This means the rules used for feature selection procedure in DG-TSK is much more than its two counterparts, which results in big training burden. P-FRB based TSK gives the performance improvement as well as the need to further improve. In our experiments,

the maximum number of the initial rules is set to 300. A direct way to lower the computation burden is reducing the maximum number of the initial rules. But this may reduce the performance as well.

5. Conclusion

In this paper, we propose a fuzzy model called DG-TSK in which double groups of gate functions are embedded for feature selection and rule extraction. Note that the feature selection and rule extraction procedures are simultaneously conducted in DG-TSK. In order to do this, a new gate function is introduced whose advantage is that it is suitable for embedding in the antecedents as well as the consequents to distinguish the important features and rules, respectively. Since this gate function is shaped like an M, it is named as M-gate. In DG-TSK, one group of M-gates are embedded into the antecedents on each feature to modulate the membership values for feature selection, while the other group of M-gates are assigned to each rule to tune the consequent parameters for rule extraction. For purpose of selecting features and extracting rules at the same time, an FRB called P-FRB that contains abundant fuzzy rules and does not grow exponentially with increasing feature dimensions is proposed. Based on the data points involved in the dataset, each one is regarded as the center of one rule and its desire label encoded by one-hot encoding is used to initialize the consequents. P-FRB based TSK has good classification performance as soon as it is initialized, which is verified in our experiments. Therefore, the proposed DG-TSK method begins with a high-quality FRB, which is of benefit to the feature selection and rule extraction. In the numerical experiments, DG-TSK is tested on 18 datasets and compared with other methods. The experimental results show that our proposed method is superior to its counterparts.

In the future researches, we plan to investigate how to adaptively determine the thresholds of the feature selection and rule extraction so as to reduce the dimensionality and compact FRB as best as possible. Besides, reducing the training time of DG-TSK deserves further study. We are going to develop more efficient strategies and faster algorithms for feature selection and rule extraction approaches. It is always difficult and complicated for fuzzy systems to deal with high-dimensional data, hence this challenge is not considered in this investigation, which needs to be done. We are currently working on this and hope to communicate our findings soon.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Double Groups of Gates based Takagi-Sugeno-Kang (DG-TSK) Fuzzy System for Simultaneous Feature Selection and Rule Extraction”.

Data availability

Data will be made available on request.

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